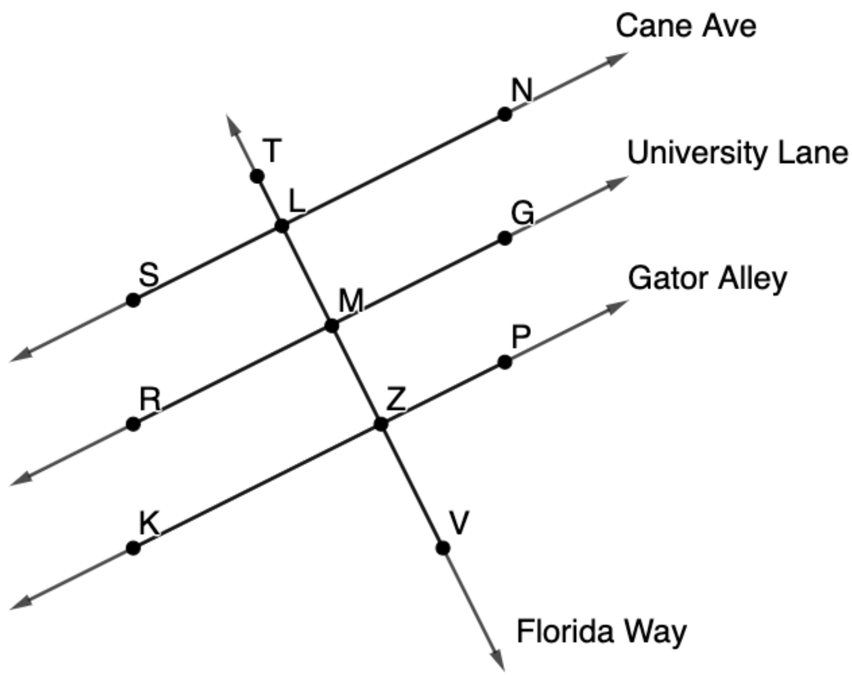
Geometry
Unit 3
Lesson 7 Practice

Name Answer Key____________
Date____________________
Period__________________

Streets Cane Ave, $\overleftrightarrow{SN}$, University Lane, $\overleftrightarrow{RG}$, and Gator Alley, $\overleftrightarrow{KP}$, are parallel to each other where Florida Way, $\overleftrightarrow{TV}$, crosses each of them as shown.



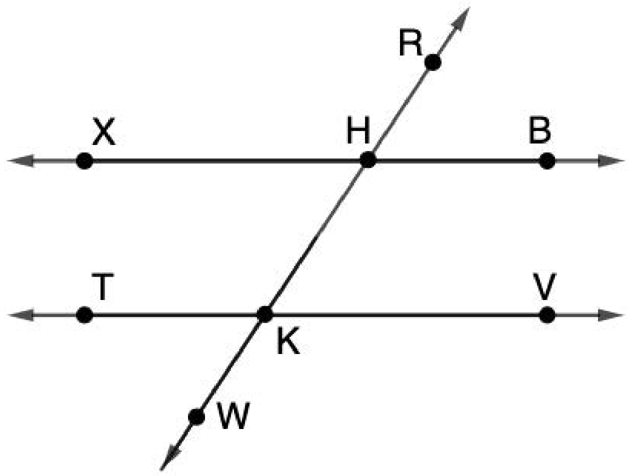
Determine the value of the variables and each angle measure.

1. $m\angle LMG = (9x + 18)°$ and $m\angle PZV = (2x + 19)°$.
   $9x + 18 + 2x + 19 = 180$
   $11x + 37 = 180$
   $x = 13$
   $m\angle LMG = 9(13) + 18 = 135°$
   $m\angle PZV = 2(13) + 19 = 45°$

2. $m\angle TLS = (9x + 19)°$, $m\angle TLN = (5x - 7)°$, and $m\angle NLM = (y - 1)°$.
   $9x + 19 + 5x - 7 = 180$
   $14x + 12 = 180$
   $x = 12$
   $m\angle TLS = 9(12) + 19 = 127°$
   $m\angle TLN = 5(12) - 7 = 53°$
   $m\angle NLM = 127° = y - 1$
   $128 = y$

3. $m\angle LMR = (9x - 7)°$, $m\angle LMG = (y + 16)°$, and $m\angle GMZ = (7x + 19)°$.
   $9x - 7 = 7x + 19$
   $2x = 26$
   $x = 13$
   $m\angle LMR = 9(13) - 7 = 110°$
   $m\angle GMZ = 7(13) + 19 = 110°$
   $m\angle LMR + m\angle LMG = 180°$
   $110 + y + 16 = 180$
   $y = 54$
   $m\angle LMG = 54 + 16 = 70°$

Lines $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$ and $\overleftrightarrow{RW}$ are shown where $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$.



Determine the value of the variables and each angle measure.

4. $m\angle TKW = (2y - 5)°$, $m\angle HKV = (9x - 19)°$, and $m\angle RHB = (8x - 11)°$.

$9x - 19 = 8x - 11$
$x = 8$

$m\angle HKV = 9(8) - 19 = 53°$
$m\angle RHB = 8(8) - 11 = 53°$
$m\angle TKW = 53°$

$2y - 5 = 53$
$y = 29$

5. $m\angle RHB = (4x + 19)°$ and $m\angle XHK = (8x + 7)°$.

$4x + 19 = 8x + 7$
$x = 3$

$m\angle RHB = 4(3) + 19 = 31°$
$m\angle XHK = 8(3) + 7 = 31°$

6. $m\angle HKV = (x - 5)°$ and $m\angle BHK = (2x + 14)°$.

$x - 5 + 2x + 14 = 180$
$3x + 9 = 180$
$x = 57$

$m\angle HKV = 57 - 5 = 52°$
$m\angle BHK = 2(57) + 14 = 128°$

7. $m\angle KHX = (6x - 5)°$ and $m\angle WKT = 101°$.

$6x - 5 = 101$
$x = 17\frac{2}{3}$

$m\angle KHX = 6\left(17\frac{2}{3}\right) - 5 = 101°$